Algebras Determined by a Braided-Symmetric Product

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ABSTRACT
An algebra is zero product determined when every bilinear map that vanishes on the pairs with zero product factors through the product. We replace the product by any bilinear map μ and ask the same question. The answer is a quotient criterion that needs no surjectivity hypothesis. For the symmetrized products of three braidings on a commutative algebra, μ has the form $f(xy)$ with f linear. The same holds on any algebra when the twist reverses products, as the transpose does on matrices. Every unital zero product determined algebra is determined by every such μ. The converse holds when f is injective, so the identity braiding gives nothing new in characteristic other than 2. The converse fails for the twisted braiding, because the dual numbers are determined by their twisted product but are not zero product determined.

KEYWORDS
zero product determined algebra; braided-symmetric product; Jordan product; dual numbers

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1. Introduction

This section states the question and the answers. Let K be a field and let A be a unital associative K-algebra. Brešar defined the zero product determined (zpd) algebras [1, Definition 1.1], and the zero Jordan product determined algebras [1, Definition 3.14]. Let $m\colon A \otimes A \to A$ be the linear map with $m(x \otimes y) = xy$. In this paper a *braiding* is a linear map σ of $A \otimes A$. Its symmetrized product is $\mu_\sigma(x, y) = m(x \otimes y + \sigma(x \otimes y))$. We use three braidings. The first is the identity. The second is the swap $x \otimes y \mapsto y \otimes x$. The third is the *twisted* swap $x \otimes y \mapsto \alpha(y) \otimes \alpha(x)$, for a linear map $\alpha\colon A \to A$. Their symmetrized products are $2xy$, $xy + yx$ and $xy + \alpha(y)\alpha(x)$. This paper asks which algebras are determined by each of these products, in the sense of Brešar.

We treat a general bilinear map μ. Section 2 recalls the definitions and the equivalent forms of Brešar and contains no new results. Section 3 proves that A is determined by μ if and only if a quotient map is injective (Proposition 3.2). Section 4 treats products of the form $\mu(x, y) = f(xy)$. Theorem 4.1 shows that every unital zpd algebra is determined by every such μ, and Proposition 4.2 proves the converse for injective f. The identity braiding has this form. The twisted braiding has it when A is commutative

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(Corollary 4.4) or when α reverses products (Corollary 4.5). Section 5 shows that the converse fails for non-injective f. Theorem 5.1 gives a sufficient condition, and Example 5.3 applies it to the dual numbers. By Corollary 4.5, the matrix algebra with the transpose is determined by its twisted product. Question 5.4 asks the same for an automorphism.

2. Preliminaries

This section recalls the notions of Brešar that we need, and none of it is new. We recall two definitions and then the equivalent forms of the first.

Definition 2.1 ([1, Definition 1.1])**.** A nonassociative algebra A over a field F is said to be *zero product determined* if, for every bilinear functional $\varphi\colon A\times A\to F$ satisfying $\varphi(x,y)=0$ whenever $x,y\in A$ are such that $xy=0$, there exists a linear functional τ on A such that $\varphi(x,y)=\tau(xy)$ for all $x,y\in A$.

We abbreviate *zero product determined* to zpd, as Brešar does.

Definition 2.2 ([1, Definition 3.14])**.** An associative algebra A is *zero Jordan product determined* if the Jordan algebra A^+ is zero product determined.

Here A^+ has the Jordan product $x\circ y=xy+yx$ in place of xy.
The next result gives equivalent forms of the first definition.

Proposition 2.3 ([1, Proposition 1.3])**.** *Let A be a nonassociative algebra over a field F. The following conditions are equivalent. (i) A is zpd. (ii) If a bilinear functional $\varphi\colon A\times A\to F$ satisfies $\varphi(x,y)=0$ whenever $xy=0$, then $\sum_i\varphi(x_i,y_i)=0$ whenever $\sum_i x_iy_i=0$. (iii) If $\sum_i z_iw_i=0$, then $\sum_i z_i\otimes w_i$ lies in $\operatorname{span}\{x\otimes y\in A\otimes A\mid xy=0\}$. (iv) For every vector space X over F and every bilinear map $\varphi\colon A\times A\to X$ with $\varphi(x,y)=0$ whenever $xy=0$, there is a linear map $T\colon A\to X$ with $\varphi(x,y)=T(xy)$ for all $x,y\in A$.*

3. Determined by a bilinear map

This section gives the definition and the criterion that later proofs use. Throughout, A is a unital associative algebra over a field K, and $\mu\colon A\times A\to A$ is K-bilinear.

Definition 3.1. The algebra A is *determined by μ* if, for every vector space V and every bilinear map $\varphi\colon A\times A\to V$ with $\varphi(x,y)=0$ whenever $\mu(x,y)=0$, there is a linear map $\psi\colon A\to V$ with $\varphi(x,y)=\psi(\mu(x,y))$ for all $x,y\in A$.

For $\mu(x,y)=xy$ and $V=K$ this is the definition of a zpd algebra, and Brešar [1, Proposition 1.3(iv)] shows that allowing any V gives the same class. For $\mu=\circ$ it is the zJpd property of Definition 2.2. The same proposition, applied to A^+, shows that allowing any V gives the same class.

The criterion uses the linear map $\bar\mu\colon A\otimes A\to A$ with $\bar\mu(x\otimes y)=\mu(x,y)$. Let $\mathcal{Z}_\mu$ be the subspace of $A\otimes A$ spanned by the tensors $x\otimes y$ with $\mu(x,y)=0$. Then $\mathcal{Z}_\mu\subseteq\ker\bar\mu$, so $\bar\mu$ induces a linear map $q_\mu\colon(A\otimes A)/\mathcal{Z}_\mu\to A$.

Proposition 3.2. *The algebra A is determined by μ if and only if q_μ is injective, that*

is, if and only if $\mathcal{Z}_\mu = \ker \bar{\mu}$.

Proof. Suppose first that A is determined by μ. Apply Definition 3.1 to the map $\varphi(x, y) = x \otimes y + \mathcal{Z}_\mu$ with values in $V = (A \otimes A)/\mathcal{Z}_\mu$. It gives ψ with $\psi(\mu(x, y)) = x \otimes y + \mathcal{Z}_\mu$. By linearity $\psi(\bar{\mu}(t)) = t + \mathcal{Z}_\mu$ for every $t \in A \otimes A$. So $\bar{\mu}(t) = 0$ implies $t \in \mathcal{Z}_\mu$.

Conversely, suppose that q_μ is injective, and let φ be as in Definition 3.1. The linear map $\bar{\varphi}\colon A \otimes A \to V$ vanishes on $\mathcal{Z}_\mu$, so it induces $\hat{\varphi}$ on the quotient. Over a field an injective linear map has a linear left inverse g with $g \circ q_\mu = \mathrm{id}$. Put $\psi = \hat{\varphi} \circ g$. Then $\psi(\mu(x, y)) = \hat{\varphi}(g(q_\mu(x \otimes y + \mathcal{Z}_\mu))) = \varphi(x, y)$. □

For $\mu(x, y) = xy$ the proposition is the equivalence of (i) and (iii) in Proposition 2.3. There q_μ is onto because $x = x \cdot 1$, and the proof can invert q_μ. A symmetrized product need not be onto, and the left inverse replaces the inverse.

4. Products of the form $f(xy)$

This section treats the case $\mu(x, y) = f(xy)$ with $f\colon A \to A$ linear, which includes two of the three braidings.

Theorem 4.1. *Let A be zpd, let $f\colon A \to A$ be linear, and let $\mu(x, y) = f(xy)$. Then A is determined by μ.*

Proof. Let φ vanish on the pairs with $\mu(x, y) = 0$. If $xy = 0$ then $\mu(x, y) = 0$, so φ vanishes on zero products. Since A is zpd, there is a linear ψ_0 with $\varphi(x, y) = \psi_0(xy)$ by Proposition 2.3(iv). Let $b \in \ker f$. Then $\mu(b, 1) = f(b) = 0$, so $\psi_0(b) = \varphi(b, 1) = 0$. Hence $\ker f \subseteq \ker \psi_0$, and ψ_0 induces a linear map on $A/\ker f$, which is isomorphic to the image of f. This gives a linear map on the image of f, and we extend it to a linear map ψ on A. Then $\psi(f(z)) = \psi_0(z)$ for every z, so $\varphi(x, y) = \psi(\mu(x, y))$. □

The next proposition gives the converse for injective f.

Proposition 4.2. *Let $f\colon A \to A$ be linear and injective, and let $\mu(x, y) = f(xy)$. If A is determined by μ, then A is zpd.*

Proof. Since f is injective, $\mu(x, y) = 0$ if and only if $xy = 0$. Let φ vanish on zero products. Then φ vanishes on the pairs with $\mu(x, y) = 0$, and we get ψ with $\varphi(x, y) = \psi(f(xy))$. So $\varphi(x, y) = (\psi \circ f)(xy)$. □

The identity braiding gives $\mu(x, y) = xy + xy = 2xy$, so we take $f = 2\,\mathrm{id}$.

Corollary 4.3. *Let $\mu(x, y) = 2xy$.*

(i) If A is zpd, then A is determined by μ.
(ii) If $\operatorname{char} K \neq 2$*, then A is determined by μ if and only if A is zpd.*

Proof. Part (i) is Theorem 4.1 with $f = 2\,\mathrm{id}$. If $\operatorname{char} K \neq 2$ this f is injective, and Proposition 4.2 gives the other direction of (ii). □

In characteristic 2 the map μ is zero, every φ in Definition 3.1 is zero, and every algebra is determined by μ.

The standard braiding gives the Jordan product, and determination by it is the zJpd property, as noted after Definition 3.1. When $\operatorname{char} K \neq 2$ the map $q_\circ$ is onto,

because $x \circ \frac{1}{2} = x$. So Proposition 3.2 recovers the usual form of the zJpd criterion. It is not of the form $f(xy)$ on a noncommutative algebra, and we refer to Brešar [1, Section 3.3] for its theory.

The twisted braiding has the form $f(xy)$ when A is commutative. Let α be an algebra endomorphism of A. Then $\alpha(y)\alpha(x) = \alpha(yx) = \alpha(xy)$, so $xy+\alpha(y)\alpha(x) = (\mathrm{id}+\alpha)(xy)$.

Corollary 4.4. *Let A be commutative and zpd, and let α be an algebra endomorphism of A. Then A is determined by $\mu_\alpha(x, y) = xy + \alpha(y)\alpha(x)$.*

Proof. Apply Theorem 4.1 with $f = \mathrm{id} + \alpha$. □

The same holds on a noncommutative algebra when α reverses products.

Corollary 4.5. *Let A be zpd, and let $\alpha\colon A \to A$ be linear with $\alpha(xy) = \alpha(y)\alpha(x)$ for all x, y. Then A is determined by $\mu_\alpha(x, y) = xy + \alpha(y)\alpha(x)$.*

Proof. Here $\alpha(y)\alpha(x) = \alpha(xy)$, so $\mu_\alpha(x, y) = (\mathrm{id} + \alpha)(xy)$. Apply Theorem 4.1 with $f = \mathrm{id} + \alpha$. □

We use a result of Brešar on matrix algebras.

For a unital algebra A_1, a unital algebra A_2 and a unital (A_1, A_2)-bimodule M, the matrices $\left[\begin{smallmatrix} r_1 & m \\ 0 & r_2 \end{smallmatrix}\right]$ with $r_1 \in A_1$, $m \in M$ and $r_2 \in A_2$ form an algebra under the usual matrix operations, which Brešar denotes by $\mathrm{Tri}(A_1, M, A_2)$ [1, Section 2.1].

Corollary 4.6 ([1, Corollary 2.17])**.** *The following algebras are zpd: (a) The algebra $M_n(B)$, where $n \geq 2$ and B is any unital algebra. (b) The algebra $\mathrm{Tri}(A_1, M, A_2)$, where A_1 and A_2 are unital algebras generated by idempotents and M is a unital (A_1, A_2)-bimodule.*

By Corollary 4.6(a) with $B = K$, the matrix algebra $M_n(K)$ is zpd for $n \geq 2$. The transpose reverses products. So $M_n(K)$ is determined by $xy + y^{\mathsf{T}}x^{\mathsf{T}}$.

When α is an involution and $\mathrm{char}\, K \neq 2$, the map $\mathrm{id} + \alpha$ is twice the projection onto the fixed points of α. It is injective only for $\alpha = \mathrm{id}$, so Proposition 4.2 gives no converse. The next section shows that the converse is false.

5. Determined but not zpd

This section gives a sufficient condition that does not assume zpd, and applies it to the dual numbers.

Theorem 5.1. *Let $f\colon A \to A$ be linear, let $N = \ker f$, and let $\mu(x, y) = f(xy)$. Suppose that $f(1) \neq 0$, that $bn, nb \in N$ for all $n \in N$ and $b \in A$, and that there is a linear functional c on A with $x - c(x)1 \in N$ for every x. Then A is determined by μ.*

Proof. Let φ vanish on the pairs with $\mu(x, y) = 0$. If $n \in N$ then ny and xn lie in N, so $\mu(n, y) = \mu(x, n) = 0$ and $\varphi(n, y) = \varphi(x, n) = 0$. Write $x = (x - c(x)1) + c(x)1$ and do the same for y. Bilinearity gives $\varphi(x, y) = c(x)c(y)\varphi(1, 1)$. The same splitting gives $f(y) = c(y)f(1)$, and then $f(xy) = c(x)f(y) = c(x)c(y)f(1)$. Since $f(1) \neq 0$, the map $a \mapsto af(1)$ from K to A has a linear left inverse g. Put $\psi(z) = g(z)\varphi(1, 1)$. Then $\psi(f(xy)) = c(x)c(y)\varphi(1, 1) = \varphi(x, y)$. □

In words, the hypotheses say that N is a two-sided ideal of codimension 1 that does

not contain 1. The next example meets them.

We also use the characterization of finite-dimensional zpd algebras by Brešar [1, Theorem 2.27].

Theorem 5.2 ([1, Theorem 2.27])**.** *A finite-dimensional unital algebra is zpd if and only if it is generated by idempotents.*

The next example uses this characterization to show that the class of algebras determined by the twisted product is strictly larger.

Example 5.3 (Dual numbers)**.** Let $\operatorname{char} K \neq 2$ and let $A = K[\varepsilon]/(\varepsilon^2)$. The map $\alpha(a+b\varepsilon) = a - b\varepsilon$ is an involutive automorphism. Since A is commutative, $\mu_\alpha(x,y) = f(xy)$ with $f = \mathrm{id}+\alpha$, and $f(a+b\varepsilon) = 2a$. So $N = K\varepsilon$, which is an ideal, $f(1) = 2 \neq 0$, and $c(a+b\varepsilon) = a$ satisfies the last hypothesis. By Theorem 5.1, A is determined by μ_α. The only idempotents of A are 0 and 1, so A is not generated by idempotents. By Theorem 5.2, A is not zpd.

So the twisted braiding determines a class of algebras strictly larger than the zpd ones, even among commutative algebras. When α is an automorphism of a noncommutative algebra, the twisted product $xy + \alpha(yx)$ is not of the form $f(xy)$, and Theorems 4.1 and 5.1 do not apply.

Question 5.4. Let $A = M_2(\mathbb{F}_3)$ and let α be conjugation by $\operatorname{diag}(1,-1)$, an involutive automorphism. Is A determined by $\mu_\alpha(x,y) = xy + \alpha(yx)$? More generally, which noncommutative algebras with an involutive automorphism are determined by their twisted product?

By Proposition 3.2, the first part of the question is a rank computation in a space of dimension 16.

References

[1] Brešar, M.: Zero Product Determined Algebras. Frontiers in Mathematics. Birkhäuser, Cham (2021).